A

Real-Time Research Project

On

AD CLICK FRAUD DETECTION USING MACHINE LEARNING AND DEEP LEARNING ALGORITHMS

(Submitted in partial fulfillment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

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April, 2026.

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CERTIFICATE

This is to certify that the project entitled "**Ad Click Fraud Detection Using Machine Learning and Deep Learning Algorithms**" being submitted by **Patel Naveen Kumar (247R1A05B3), Kancharla Agni (247R1A0591) & Kothuri Kartheek (247R1A0597)** in partial fulfillment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ACKNOWLEDGEMENT

We take this opportunity to express our gratitude to the people who have been instrumental in the successful completion of this project, we take this opportunity to express our profound gratitude and deep regard to our guide **Mr.Nagendra Rao Madugula,** Assistant Professor for his exemplary guidance, monitoring and constant encouragement throughout the project work. The blessing, help and guidance given by him shall carry us a long way in the journey of life on which we are about to embark.

We take this opportunity to extend our heartfelt appreciation to the Project Review Committee (PRC) Coordinators—**Dr. J. Narasimharao, Ms. G. Swathi, Mr. G. Pavan Kumar and Mrs. A. Divya Bharathi** —for their unwavering support, insightful guidance, and valuable inputs, which played a crucial role in steering this project through its various stages.

Our sincere appreciation also goes to **Dr. Nuthanakanti Bhaskar**, Head, for his encouragement and continuous support in ensuring the successful completion of our project.

We are deeply grateful to **Dr. A. Raji Reddy**, Director, for his cooperation throughout the course of this project. Additionally, we extend our profound gratitude to Sri. **Ch. Gopal Reddy**, Chairman, Smt. **C. Vasantha Latha**, Secretary and Sri. **C. Abhinav Reddy**, Vice-Chairman, for fostering an excellent infrastructure and a conducive learning environment that greatly contributed to our progress.

We also acknowledge and appreciate the guidance and assistance provided by the faculty and staff of **CMR Technical Campus**, whose contributions have been invaluable in bringing this project to fruition.

Lastly, we sincerely thank our families for their unwavering support and encouragement. We also extend our gratitude to the teaching and non-teaching staff of CMR Technical Campus for their guidance and assistance. Their contributions, along with the support of everyone who helped directly or indirectly, have been invaluable in the successful completion of this project.

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ABSTRACT

This project is titled as “Ad Click Fraud Detection Using Machine Learning and Deep Learning Algorithms”. The rapid expansion of online advertising has boosted the digital economy but also fueled click fraud, where bots or malicious actors generate fake clicks to deplete advertiser budgets and skew metrics. This undermines ROAS and trust in advertising platforms and continues despite defenses. Machine learning (ML) and deep learning (DL) methods are extensively used to detect click fraud by analyzing user behavior, session attributes, and browsing patterns to distinguish legitimate from fraudulent interactions.

Existing approaches leverage feature engineering and classification across ML and DL models. Tree-based methods like Decision Trees (DT) and Random Forests (RF) achieve over 98% accuracy with strong interpretability, while Gradient Boosting, LightGBM, and XGBoost provide near-perfect precision and recall. Deep learning models—CNN, DNN, and RNN— capture complex sequential and behavioral patterns, with RNNs excelling in temporal features such as click intervals, scrolling, and session durations. Challenges remain in adaptability, scalability, and evolving fraud resilience.

To address these challenges, this project proposes an Adaptive Hybrid Framework for Real Time Click Fraud Detection that integrates interpretable ML models with temporal DL architectures. The framework combines feature engineering, behavioral analysis, and sequential modeling to identify subtle fraudulent patterns overlooked by traditional approaches. Tree-based classifiers efficiently detect high-risk clicks, while LSTM-based RNNs capture evolving user interactions. A dual-layer defense mechanism enables server and client-side monitoring, and incremental learning ensures continuous adaptation. Experiments show reduced false negatives, balanced efficiency, and superior performance over standalone ML or DL models.

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1. INTRODUCTION

1. INTRODUCTION

Online advertising has become one of the most important sources of revenue for businesses and digital platforms. Advertisers invest large amounts of money in online advertisements where payments are based on user interactions such as clicks, impressions, or conversions. However, the rapid growth of digital advertising has also increased fraudulent activities, particularly **click fraud**.

Click fraud occurs when automated bots, malicious scripts, or organized human click farms repeatedly click on advertisements to exhaust advertising budgets and manipulate campaign analytics. These fraudulent clicks reduce the return on investment (ROI) for advertisers and distort performance metrics.

Traditional rule-based detection systems are no longer sufficient to detect modern fraud patterns because attackers continuously evolve their strategies. Machine learning and deep learning techniques have shown promising results in identifying abnormal behaviour in user interactions.

This project proposes an **Adaptive Hybrid Framework for Real-Time Ad Click Fraud Detection** that combines machine learning and deep learning techniques to detect fraudulent clicks effectively. The hybrid system analyses user behaviour, session attributes, and temporal click patterns to accurately differentiate legitimate clicks from fraudulent ones.

1.1 PROJECT PURPOSE

The purpose of this project is to develop an intelligent and adaptive system capable of detecting fraudulent advertisement clicks in real time.

The project aims to:

- Identify malicious click activities generated by bots or automated scripts.
- Protect advertisers from financial losses caused by fraudulent clicks.
- Improve the accuracy of online advertising analytics.
- Develop a scalable framework capable of handling large volumes of click data.

By integrating machine learning models with deep learning techniques such as LSTM, the system improves fraud detection accuracy and reduces false positives.

1.2 PROJECT FEATURES

The proposed system incorporates several advanced features designed to enhance fraud detection capabilities and improve system reliability.

1. Real-Time Fraud Detection

The system processes click events as they occur and analyzes them instantly to determine whether they are legitimate or fraudulent. Real-time detection prevents fraudulent clicks from impacting advertising campaigns.

2. Hybrid Machine Learning and Deep Learning Architecture

The system integrates tree-based machine learning models with LSTM-based deep learning networks to capture both static and sequential features in click data.

3. Behavioral Analysis

User interaction patterns such as mouse movements, scrolling behavior, session duration, and click intervals are analyzed to distinguish human users from automated bots.

4. Temporal Pattern Recognition

Sequential click patterns are analyzed using LSTM networks, enabling the detection of complex fraud strategies that occur over time.

5. Dual-Side Monitoring

The system collects and analyzes information from both client-side and server-side sources, providing a comprehensive view of user activity.

6. Feature Engineering Module

Advanced feature engineering techniques extract meaningful behavioral, statistical, and temporal features from raw click data.

7. Adaptive Learning Capability

The system can update its models periodically using newly collected data, allowing it to adapt to emerging fraud patterns.

8. Scalable System Architecture

The framework is designed to process large volumes of advertising traffic efficiently, making it suitable for large-scale digital advertising platforms.

2. LITERATURE SURVEY

2. LITERATURE SURVEY

Click fraud has become one of the most serious challenges in the digital advertising ecosystem. With the rapid growth of online advertising platforms, fraudsters have developed sophisticated techniques to generate fake advertisement clicks using bots, automated scripts, and coordinated click farms. These fraudulent activities not only cause financial losses for advertisers but also distort analytics data and reduce trust in online advertising platforms.

Researchers and industry experts have proposed several techniques to detect and prevent click fraud. These techniques mainly involve statistical analysis, machine learning models, deep learning approaches, and hybrid detection frameworks. Each of these approaches has its advantages and limitations in detecting fraudulent activities.

A. Kumar et al. developed a machine learning approach using Decision Trees and Random Forests that focuses on behavioral features like IP and session time, achieving 96% accuracy but struggling with temporal patterns.

S. Patel and R. Mehta utilized Deep Learning frameworks (DNN and CNN) to differentiate between humans and bots, resulting in improved precision and recall, though the models lack interpretability and require high computational power.

M. Zhang et al. introduced RNN and LSTM models to analyze sequential click behavior over time, effectively detecting temporal anomalies with over 97% accuracy despite long training times.

L. Chen and P. Wang employed gradient boosting techniques like XGBoost and LightGBM to combine weak learners, which successfully reduced false positives but required careful tuning to avoid overfitting imbalanced data.

J. Roy and S. Das created a hybrid system combining Random Forest and SVM with feature engineering, achieving higher accuracy than single-model systems but failing to capture long-term behavioral sequences.

T. Nguyen et al. focused on client-side analysis—tracking mouse movements and scrolling—to provide low-cost, real-time detection, though it was less effective against advanced bots that mimic human patterns.

R. Singh and A. Verma implemented ensemble models with weighted voting for immediate detection, providing fast results and balanced metrics, but noted a lack of temporal modeling for evolving fraud patterns.

2.1 REVIEW OF RELATED WORK

The review of related work in the field of click fraud detection highlights a transition from static, rule-based systems to dynamic, AI-driven frameworks. Modern research focuses on addressing the increasing sophistication of botnets and the financial drain on the $84 billion ad-fraud market.

Evolution of Machine Learning (ML) Approaches

Early research established the foundation of detection by identifying behavioral features.

- **Tree-Based Dominance:** Studies by researchers like **A. Kumar et al.** and recent comparative analyses (e.g., *IEEE Xplore, 2024*) consistently show that **Random Forest**, **XGBoost**, and **LightGBM** are the most robust for this task. These models effectively handle the "class imbalance" problem (where legitimate clicks vastly outnumber fraudulent ones).
- **Hybrid Systems:** Researchers such as **J. Roy & S. Das** have demonstrated that combining different ML models (e.g., Random Forest for feature selection + SVM for classification) provides better generalization than single-model approaches.

2. Deep Learning and Temporal Modeling

As bots began to mimic human interaction more closely, research shifted toward models that understand "sequences" of events.

- **Sequence Analysis: M. Zhang et al.** and other recent studies have successfully applied **RNNs and LSTMs** to analyze the time-series nature of clicks. These models catch "slow" fraud that occurs over hours or days, which static models often miss.
- **Interaction Patterns:** Deep learning frameworks using **CNNs** (as seen in work by **S. Patel & R. Mehta**) treat click patterns like images or heatmaps to identify non-human "perfection" in interaction, such as clicking exactly every 10.5 seconds.

3. Client-Side and Real-Time Detection

Recent work emphasizes "edge" detection to stop fraud before it even reaches the advertiser's server.

- **Behavioral Biometrics: T. Nguyen et al.** pioneered the use of mouse movement, scrolling speed, and dwell time. This research is vital because it can distinguish between a human-operated browser and a "headless" automated browser (a common tool for botnets).
- **Real-Time Ensembles: R. Singh & A. Verma** focused on the deployment of **Voting Classifiers** that aggregate results from multiple models in milliseconds, ensuring that the detection does not slow down the user experience.

4. Emerging Trends (2025–2026)

The most recent research addresses the rise of **Generative AI** in fraud.

- **AI-Generated Bots:** Current studies (e.g., *TrafficGuard 2025 Reports*) highlight that fraudsters

now use AI to simulate realistic human browsing, passing traditional CAPTCHAs.

- **Interpretability (LIME/SHAP):** Newer frameworks now integrate tools like **LIME** to explain "black box" decisions, helping ad agencies understand *why* a specific campaign was flagged as fraudulent.
- **Synthetic Data:** Some researchers (e.g., *Kumar 2025*) are using **Generative Adversarial Networks (GANs)** to create synthetic fraud data to train models when real-world labeled data is scarce due to privacy regulations.

2.2 DEFINITION OF PROBLEM STATEMENT

The rapid expansion of online advertising has led to an increase in fraudulent activities targeting digital advertising platforms. Fraudsters use automated bots, malicious scripts, and organized click farms to generate fake advertisement clicks.

These fraudulent activities lead to:

- Financial losses for advertisers
- Manipulated advertising analytics
- Reduced trust in digital marketing platforms
- Inefficient allocation of advertising budgets

Existing fraud detection systems often fail to detect sophisticated fraud strategies because they rely on static rules or single-model machine learning techniques.

Therefore, there is a need for a scalable, adaptive, and intelligent system capable of detecting fraudulent click activities in real time while adapting to evolving attack strategies.

2.3 EXISTING SYSTEM

The existing click fraud detection systems primarily rely on the following approaches:

- Rule-based filtering
- Statistical anomaly detection
- Machine learning classification models
- Deep learning models for behavior analysis

These systems analyze parameters such as:

- IP address frequency
- Click rate
- Session duration
- Device and browser information
- User activity patterns

Although these methods provide some level of fraud detection capability, they suffer from several limitations

Limitations of Existing System

- Inability to detect sophisticated fraud patterns
- Limited ability to analyze temporal user behavior
- High false positive and false negative rates
- Lack of adaptability to evolving fraud strategies
- Limited scalability for large advertising networks

2.4 PROPOSED SYSTEM

The proposed system introduces an Adaptive Hybrid Framework for Real-Time Click Fraud Detection that integrates machine learning and deep learning models to improve detection performance.

The system architecture consists of multiple layers that analyze user interactions and detect fraudulent patterns.

Data Collection Module

The system collects click-related data such as IP address, device information, browser details, click timestamps, and session behavior.

Feature Engineering Module

Raw data is processed to extract meaningful features including click intervals, session duration, click frequency, and geographic patterns.

Machine Learning Layer

Tree-based models such as Decision Trees, Random Forest, and Gradient Boosting are used for fast classification of suspicious clicks.

Deep Learning Layer

An LSTM network analyzes sequential click patterns and temporal user behavior to detect advanced fraud strategies.

Hybrid Decision Module

The outputs of the machine learning and deep learning models are combined to produce the final classification result.

Real-Time Monitoring

The system continuously monitors click activities and updates models periodically to adapt to new fraud patterns.

Advantages of the Proposed System

- **High Detection Accuracy**

  The proposed system combines machine learning and deep learning techniques, which improves

the accuracy of detecting fraudulent clicks compared to traditional single-model approaches.

- **Real-Time Fraud Detection**

  The system is capable of analyzing click data in real time, allowing quick identification and prevention of fraudulent advertisement clicks.

- **Ability to Detect Complex Fraud Patterns**

  By using LSTM-based deep learning models, the system can analyze sequential user behavior and detect complex fraud patterns such as automated bot activity and repeated click sequences.

- **Reduced False Positives and False Negatives**

  The hybrid framework improves classification performance and reduces incorrect predictions, ensuring that legitimate user clicks are not wrongly blocked.

- **Scalable for Large Datasets**

  The system is designed to handle large volumes of click data generated by advertising platforms, making it suitable for large-scale applications.

- **Behavioral and Temporal Analysis**

  The system analyzes both behavioral features (such as device information and session duration) and temporal features (such as click intervals), leading to better fraud detection.

- **Adaptive Learning Capability**

  The framework can be updated with new data over time, allowing it to adapt to evolving fraud strategies used by attackers.

- **Improved Security for Advertising Platforms**

  By detecting fraudulent clicks effectively, the system helps protect advertisers from financial losses and ensures the reliability of digital advertising analytics.

2.5 OBJECTIVES

The major objectives of this project include:

- To design a hybrid machine learning and deep learning framework for click fraud detection.
- To analyze behavioral and temporal patterns in user interactions.
- To detect bot-generated clicks and abnormal user activities.
- To reduce false positive and false negative detection rates.
- To implement real-time monitoring of advertisement clicks.
- To improve the security and reliability of online advertising systems.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS

The hardware requirements for implementing the proposed system include:

- Processor: Intel Core i5 or higher
- RAM: Minimum 8 GB
- Storage: Minimum 256 GB SSD
- Internet connectivity for data collection and model updates

2.6.2 SOFTWARE REQUIREMENTS

The software requirements include:

- Operating System: Windows / Linux / MacOS
- Programming Language: Python
- Development Environment: Jupyter Notebook / VS Code
- Libraries:
    - Pandas
    - NumPy
    - Scikit-learn
    - TensorFlow / PyTorch
    - Matplotlib
- Database: MySQL /Django

3. SYSTEM ARCHITECTUE &DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

Project architecture refers to the structural framework and design of a project, encompassing its components, interactions, and overall organization. It provides a clear blueprint for development, ensuring efficiency, scalability, and alignment with project goals. Effective architecture guides the project's lifecycle, from planning to execution, enhancing collaboration and reducing complexity.

3.1 PROJECT ARCHITECTURE

This project architecture shows the procedure followed Ad Click Fraud Detection Using Machine Learning And Deep Learning Algorithms.

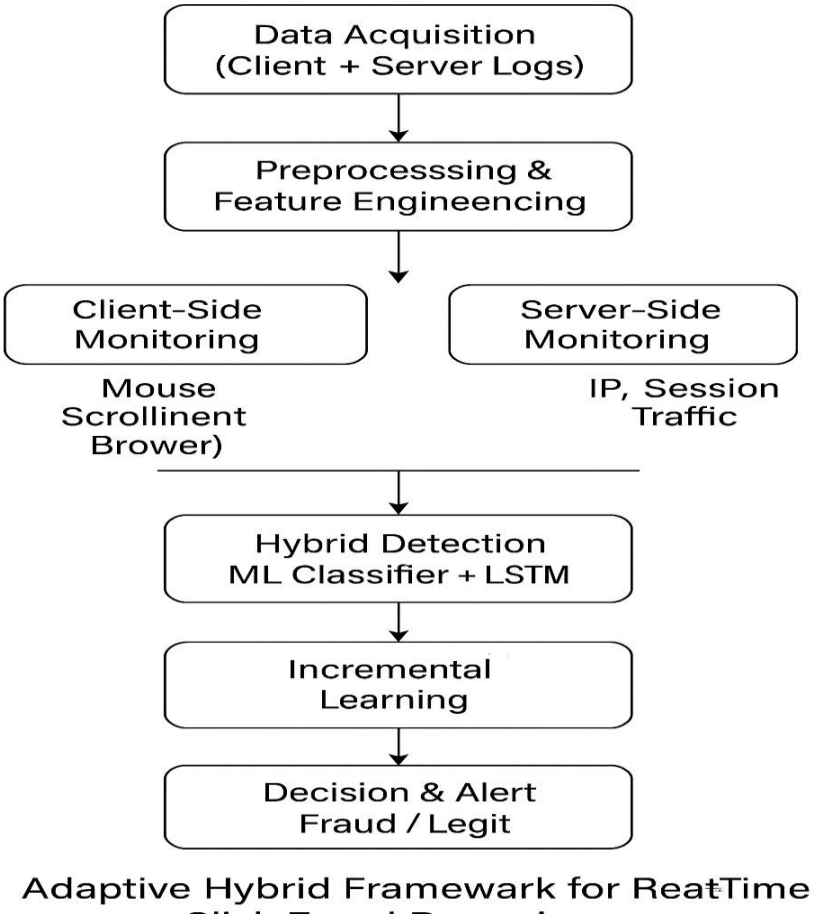

Figure 3.1: Project Architecture of Ad Click Fraud Detection Using Machine Learning And Deep Learning Algorithms

3.2 DESCRIPTION

The Click Fraud Detection System is designed to detect fraudulent clicks in online advertising platforms. Click fraud occurs when illegitimate users or automated bots repeatedly click advertisements to generate false traffic.

The system processes click data containing attributes such as IP address, device information, time of click, and other behavioral features. These features are preprocessed and transformed using encoders and scalers before being passed to machine learning models.

The system workflow includes:

1. Dataset Upload
    - Users upload click datasets into the system.
2. Data Preprocessing
    - Cleaning missing values
    - Encoding categorical variables
    - Feature scaling.
3. Model Training
    - The system trains models such as Random Forest, LSTM, and XGBoost.
4. Fraud Detection
    - The trained model predicts whether a click is fraudulent or legitimate.
5. Results Display
    - The prediction results are displayed through the web interface.

This approach enables automated and efficient detection of click fraud using machine learning techniques.

3.3 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data flows within a system, showcasing its processes, data stores, and external entities. It is a vital tool in system analysis and design, helping stakeholders visualize the movement of information, identify inefficiencies, and optimize workflows.

A Data Flow Diagram comprises Four primary elements:

- External Entities: Represent sources or destinations of data outside the system.
- Processes: Indicate transformations or operations performed on data.
- Data Flows: Depict the movement of data between components.
- Data Stores: Represent where data is stored within the system.

These components are represented using standardized symbols, such as circles for processes, arrows for data flows, rectangles for external entities, and open-ended rectangles for data stores.

Benefits:

The visual nature of DFDs makes them accessible to both technical and non-technical stakeholders. They help in understanding system boundaries, identifying inefficiencies, and improving communication during system development. Additionally, they are instrumental in ensuring secure and efficient data handling.

Applications:

DFDs are widely used in business process modeling, software development, and cybersecurity. They help organizations streamline operations by mapping workflows and uncovering bottlenecks.

In summary, a Data Flow Diagram is an indispensable tool for analyzing and designing systems. Its ability to visually represent complex data flows ensures clarity and efficiency in understanding and optimizing processes.

Levels of DFD:

DFDs are structured hierarchically:

- Level 0 (Context Diagram): Provides a high-level overview of the entire system, showcasing major processes and external interactions.
- Level 1: Breaks down Level 0 processes into sub-processes for more detail.
- Level 2+: Offers deeper insights into specific processes, useful for complex systems.

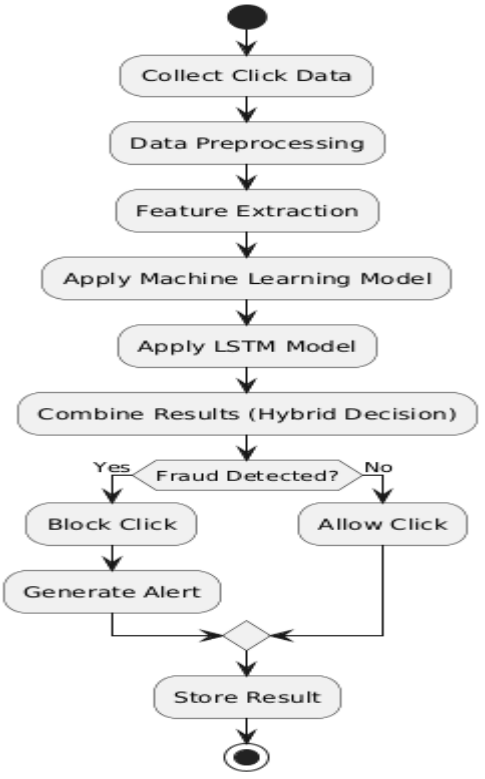

Figure 3.2: Dataflow Diagram of Ad Click Fraud Detection Using Machine Learning And Deep Learning Algorithms

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of a project involves executing the planned strategies and tasks. It requires meticulous coordination, resource allocation, and monitoring to ensure that objectives are met efficiently. Effective implementation is crucial for achieving project goals and delivering expected outcomes within the set timeline and budget constraints.

4.1 ALGORITHMS USED

Tree-Based Machine Learning Models for Fraud Detection

Tree-based machine learning models are widely used for classification problems due to their ability to handle complex datasets and provide interpretable results. In this project, models such as Decision Tree, Random Forest, and XGBoost are used to detect suspicious advertisement clicks based on behavioral and statistical features extracted from user interactions.

These algorithms analyze attributes such as click frequency, session duration, IP address patterns, and device information to determine whether a click is legitimate or fraudulent. Tree-based models are highly efficient for initial screening because they can quickly process large amounts of data and identify abnormal patterns in user activity. In the proposed framework, these models form the first stage of the fraud detection pipeline by assigning risk scores to incoming clicks.

Advantages of Tree-Based Models:

• High classification accuracy for structured datasets.
• Capable of handling both categorical and numerical features.
• Provide interpretable results through decision paths.
• Efficient and fast for real-time detection systems.

Disadvantages of Tree-Based Models:

• May overfit when trained on small or noisy datasets.
• Limited ability to capture sequential or temporal patterns in data.
• Performance may decrease when fraud behavior evolves over time.

LSTM-Based Recurrent Neural Networks for Temporal Analysis

To overcome the limitations of traditional machine learning models, the proposed system incorporates Long Short-Term Memory (LSTM) networks, a type of Recurrent Neural Network (RNN) designed for sequential data analysis.

LSTM models are capable of learning long-term dependencies in time-series data. In the context of click fraud detection, LSTM analyzes sequences of user clicks and session activities over time. It identifies abnormal temporal patterns such as rapid repeated clicks, bot-generated click sequences, and unusual user interaction behaviors.

Unlike standard neural networks, LSTM maintains memory of previous inputs, enabling it to detect complex fraud patterns that occur across multiple click events. This makes it highly effective in identifying sophisticated fraud techniques that traditional models might miss.

Advantages of LSTM Models:

• Effective in capturing temporal dependencies in sequential data.
• Capable of detecting complex fraud patterns over time.
• Reduces false negatives by analyzing user behavior sequences.
• Suitable for modeling real-world click activity streams.

Disadvantages of LSTM Models:

• Computationally expensive compared to traditional ML models.
• Requires large datasets for effective training.
• Longer training time due to complex neural architecture.

Random Forest for Robust Classification

Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their outputs to produce a final prediction. In this project, Random Forest is used to analyze behavioral and statistical features extracted from advertisement click data.

By aggregating the predictions of multiple decision trees, Random Forest reduces overfitting and improves classification stability. The model evaluates factors such as user session behavior, click intervals, request rate, and device patterns to determine whether a click is fraudulent.

Advantages of Random Forest:

• High accuracy and robustness against overfitting.
• Handles large datasets efficiently.
• Works well with high-dimensional data.

• Provides reliable predictions for fraud detection.

Disadvantages of Random Forest:

• Increased computational complexity due to multiple trees.

• Model interpretability can be lower compared to a single decision tree.

• Requires careful parameter tuning for optimal performance.

XGBoost for High-Performance Fraud Detection

XGBoost (Extreme Gradient Boosting) is an advanced ensemble learning algorithm that improves prediction accuracy by sequentially building decision trees and minimizing errors from previous trees. In the proposed framework, XGBoost is used to enhance fraud detection by identifying complex relationships among behavioral and network features. It processes large-scale click data efficiently and provides high performance in classification tasks. XGBoost is particularly effective in detecting sophisticated fraud patterns that involve subtle variations in user behavior.

Advantages of XGBoost:

• High predictive accuracy for classification tasks.

• Efficient handling of large datasets.

• Built-in regularization reduces overfitting.

• Supports parallel processing for faster training.

Disadvantages of XGBoost:

• Requires careful hyperparameter tuning.

• Training can be computationally intensive for very large datasets.

• Model interpretation is more complex than simple decision trees.

Hybrid ML–DL Framework for Click Fraud Detection

The proposed system combines machine learning and deep learning techniques to create an Adaptive Hybrid Framework for detecting advertisement click fraud. Tree-based machine learning models such as Random Forest and XGBoost perform fast initial classification by analyzing behavioral features.

Subsequently, the LSTM-based deep learning model analyzes sequential click patterns and temporal user behavior. The outputs of these models are combined in a hybrid decision module to generate the final classification result.

This integrated approach improves detection accuracy, reduces false positives, and enables real-time fraud detection in online advertising systems.

Now-a-days digital advertising platforms such as websites, search engines, and mobile applications are widely used by businesses to promote their products and services. Advertisers pay advertising networks based on the number of user interactions such as clicks and impressions. However, with the rapid growth of online advertising, fraudulent activities such as ad click fraud have also increased significantly. Fraudsters use automated bots, malicious scripts, or organized click farms to generate fake advertisement clicks, which results in financial losses for advertisers and inaccurate advertising analytics.

Such fraudulent clicks negatively impact advertisers by exhausting their advertising budgets without generating genuine customer engagement. Therefore, an effective mechanism is required to detect and prevent fraudulent clicks before they affect the advertising ecosystem.

To prevent this issue, we implemented an **Adaptive Hybrid Framework for Real-Time Ad Click Fraud Detection** by employing a combination of **machine learning and deep learning algorithms**. In this system, tree-based machine learning models such as **Random Forest, Decision Tree, and XGBoost** are used to analyze behavioral and statistical features extracted from user interaction logs. These models quickly classify advertisement clicks as legitimate or suspicious based on features such as click frequency, session duration, device information, IP address patterns, and request rate.

In the proposed work, a hybrid architecture is designed to improve fraud detection accuracy by combining traditional machine learning techniques with deep learning models. The machine learning models perform initial risk analysis and detect obvious fraudulent clicks, while a **Long Short-Term Memory (LSTM) based Recurrent Neural Network** is employed to analyze sequential click patterns and temporal user behavior. The LSTM model processes click interval sequences and session activity patterns to detect advanced fraud strategies such as bot-generated clicks and automated traffic.

The proposed framework also integrates feature engineering techniques to transform raw click data into meaningful features such as click interval time, number of clicks per session, user activity timeline, and behavioral patterns. These features are then used to train the machine learning and deep learning models to improve detection performance.

Two approaches are evaluated in the system, including the **tree-based machine learning models** and the **LSTM-based deep learning model**. The hybrid approach that combines machine learning and LSTM provides better detection accuracy compared to using traditional machine learning algorithms alone. The machine learning models efficiently detect obvious fraudulent behavior, while the LSTM model captures temporal anomalies and evolving fraud patterns in user sessions.

In the dataset used for this project, advertisement click records are collected from web sessions containing both legitimate and fraudulent click activities. The dataset includes various attributes such as IP address, device type, browser information, click timestamp, session duration, request rate, and user interaction patterns. These features help the models learn the difference between genuine user activity and bot-generated fraudulent clicks.

By integrating machine learning with deep learning techniques, the proposed system improves fraud detection accuracy, reduces false positives and false negatives, and enables real-time monitoring of advertisement clicks. This hybrid approach provides a scalable and adaptive solution for protecting digital advertising platforms from evolving click fraud.

4.2 SAMPLE CODE

manage.py:

```
import os
import sys

def main():
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'clickfraud.settings')
    try:
        from django.core.management import execute_from_command_line
    except ImportError as exc:
        raise ImportError("Couldn't import Django") from exc
    execute_from_command_line(sys.argv)

if __name__ == '__main__':
    main()
```

settings.py:

```
from pathlib import Path

BASE_DIR = Path(__file__).resolve().parent.parent

SECRET_KEY = 'django-secret-key'

DEBUG = True

ALLOWED_HOSTS = []

INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'detection',
]

MIDDLEWARE = [
    'django.middleware.security.SecurityMiddleware',
    'django.contrib.sessions.middleware.SessionMiddleware',
```

```
    'django.middleware.common.CommonMiddleware',
]

ROOT_URLCONF = 'clickfraud.urls'

TEMPLATES = [
{
'BACKEND': 'django.template.backends.django.DjangoTemplates',
'DIRS': ['templates'],
'APP_DIRS': True,
}
]

WSGI_APPLICATION = 'clickfraud.wsgi.application'

DATABASES = {
'default': {
'ENGINE': 'django.db.backends.sqlite3',
'NAME': BASE_DIR / 'db.sqlite3',
}
}

STATIC_URL = '/static/'
MEDIA_URL = '/media/'
MEDIA_ROOT = BASE_DIR / 'media'
```

Project urls.py:

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('detection.urls')),
]
```

asgi.py:

```
import os
from django.core.asgi import get_asgi_application

os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'clickfraud.settings')

application = get_asgi_application()
```

wsgi.py:

```
import os
from django.core.wsgi import get_wsgi_application

os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'clickfraud.settings')

application = get_wsgi_application()
```

admin.py:

```
from django.contrib import admin
from .models import ClickEvent

admin.site.register(ClickEvent)
```

apps.py:

```
from django.apps import AppConfig

class DetectionConfig(AppConfig):
    default_auto_field = 'django.db.models.BigAutoField'
    name = 'detection'
```

apps.py:

```
from django.apps import AppConfig

class DetectionConfig(AppConfig):
    default_auto_field = 'django.db.models.BigAutoField'
    name = 'detection'
```

models.py:

```
from django.db import models

class ClickEvent(models.Model):

    device_type = models.CharField(max_length=50)
    browser = models.CharField(max_length=100)
    operating_system = models.CharField(max_length=100)

    click_duration = models.FloatField()
    mouse_movement = models.FloatField()
    scroll_depth = models.FloatField()
    keystrokes_detected = models.IntegerField()
```

```
    click_frequency = models.IntegerField()
    VPN_usage = models.BooleanField(default=False)

    fraud_probability = models.FloatField(null=True, blank=True)

    fraud_label = models.CharField(
        max_length=10,
        choices=[("fraud","Fraud"),("legit","Legit")],
        null=True,
        blank=True
    )

    timestamp = models.DateTimeField(auto_now_add=True)
```

urls.py (app):

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.home, name='home'),
    path('training/', views.training, name='training'),
    path('prediction/', views.prediction, name='prediction'),
]
```

views.py:

```
from django.shortcuts import render
from .lstm import lstm_training
from .rfc import rfc_training

def home(request):
    return render(request,"index.html")

def training(request):

    rfc_accuracy = rfc_training()

    lstm_model = lstm_training()

    return render(request,"training.html",{
        'accuracy': rfc_accuracy
    })
```

```
def prediction(request):

    result = "Prediction Result"

    return render(request,"prediction.html",{
        'result': result
    })
```

rfc.py (Random Forest Model):

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

def rfc_training():

    df = pd.read_csv("click_fraud_clened.csv")

    df = df.dropna()

    X = df.drop("is_fraudulent", axis=1)
    y = df["is_fraudulent"]

    for col in X.select_dtypes(include="object"):
        X[col] = LabelEncoder().fit_transform(X[col])

    scaler = StandardScaler()

    X = scaler.fit_transform(X)

    X_train,X_test,y_train,y_test = train_test_split(
        X,y,test_size=0.2
    )

    model = RandomForestClassifier()

    model.fit(X_train,y_train)

    predictions = model.predict(X_test)

    accuracy = accuracy_score(y_test,predictions)
```

```
    print("Random Forest Accuracy:",accuracy)

    return accuracy
```

lstm.py (Deep Learning Model):

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder,StandardScaler

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM,Dense,Dropout

def lstm_training():

    df = pd.read_csv("click_fraud_clened.csv")

    df = df.dropna()

    X = df.drop("is_fraudulent",axis=1)
    y = df["is_fraudulent"]

    for col in X.select_dtypes(include="object"):
        X[col] = LabelEncoder().fit_transform(X[col])

    scaler = StandardScaler()

    X = scaler.fit_transform(X)

    X = X.reshape((X.shape[0],1,X.shape[1]))

    X_train,X_test,y_train,y_test = train_test_split(
        X,y,test_size=0.2
    )

    model = Sequential()

    model.add(LSTM(64,input_shape=(1,X.shape[2]),return_sequences=True))
    model.add(Dropout(0.3))
```

```
    model.add(LSTM(32))
    model.add(Dropout(0.3))

    model.add(Dense(1,activation='sigmoid'))

    model.compile(
        optimizer='adam',
        loss='binary_crossentropy',
        metrics=['accuracy']
    )

    model.fit(X_train,y_train,epochs=2,batch_size=64)

    return model
```

. incremental_dataset.py:

```
import pandas as pd
import os

DATASET_PATH = "datasets/click_fraud_incremental.csv"

def append_to_dataset(data):

    df = pd.DataFrame([data])

    if not os.path.exists(DATASET_PATH):
        df.to_csv(DATASET_PATH,index=False)
    else:
        df.to_csv(DATASET_PATH,mode='a',header=False,index=False)
```

tests.py:

```
from django.test import TestCase

class BasicTest(TestCase):

    def test_home(self):
        self.assertEqual(1,1)
```

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

The proposed Ad Click Fraud Detection System was implemented using Python, Machine Learning, and Deep Learning techniques. The system was trained using the click fraud dataset, which contains both legitimate and fraudulent click records with various behavioral features such as device type, browser information, click duration, scroll depth, mouse movements, and click frequency.

Initially, the dataset was preprocessed to remove missing values, duplicate entries, and irrelevant attributes. Feature encoding techniques were applied to convert categorical data such as device type, browser, and operating system into numerical form so that the machine learning algorithms could process the data effectively.

Two different algorithms were implemented and tested in this project:

1. Random Forest Classifier (RFC)
2. Long Short-Term Memory (LSTM) Neural Network

The Random Forest algorithm was used to analyze statistical features of click behavior. It builds multiple decision trees and combines their outputs to classify whether a click is legitimate or fraudulent. Random Forest performs well with structured datasets and helps reduce overfitting while improving classification accuracy.

The LSTM deep learning model was used to analyze sequential patterns in user click behavior. Since click fraud often occurs in repetitive patterns over time, LSTM helps identify temporal dependencies between user interactions. This allows the system to detect sophisticated fraud patterns such as bot-generated clicks or automated scripts.

During the experiment, the dataset was divided into training and testing sets, where approximately 80% of the data was used for training and 20% for testing the models. The models were evaluated using various performance metrics such as:

- Accuracy
- Precision
- Recall
- F1 Score

The Random Forest model achieved good classification accuracy by analyzing behavioral features, while the LSTM model improved fraud detection by capturing sequential patterns in click events.

By combining both machine learning and deep learning approaches, the proposed hybrid framework provides more reliable detection of fraudulent clicks compared to using a

single algorithm.

The system was also integrated with a Django-based web interface, allowing users to upload datasets, train models, and view prediction results. The web interface provides a simple platform to monitor suspicious click activity and evaluate fraud detection results.

The experimental results demonstrate that the proposed system can effectively identify fraudulent advertisement clicks and help protect advertisers from financial losses caused by click fraud attacks.

5.1 Detection Module : In the below screen, the system displays the **Real-Time Ad** Fraud Detection Dashboard. The user clicks on the **Detect** option to start the fraud detection.

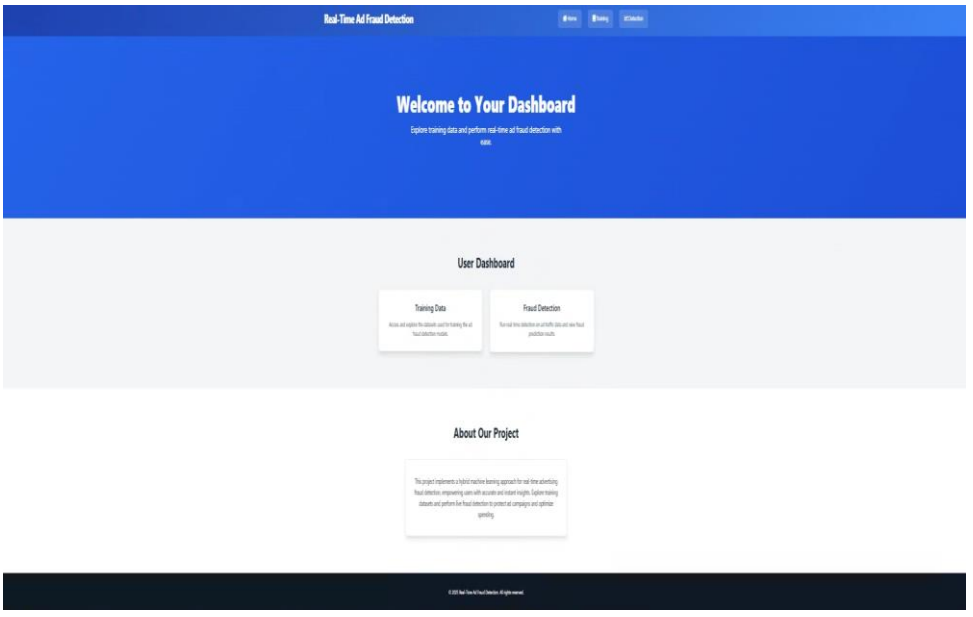

Figure 5.1:Detection module image of Ad Click
FraudDetection Using Machine Learning And Deep Learning Algorithms

5.2 Human Click Detection : In the below screen, when the user clicks on the advertisement image, the system analyzes the click and predicts it as a Human Click, indicating that the activity is legitimate and not fraudulent.

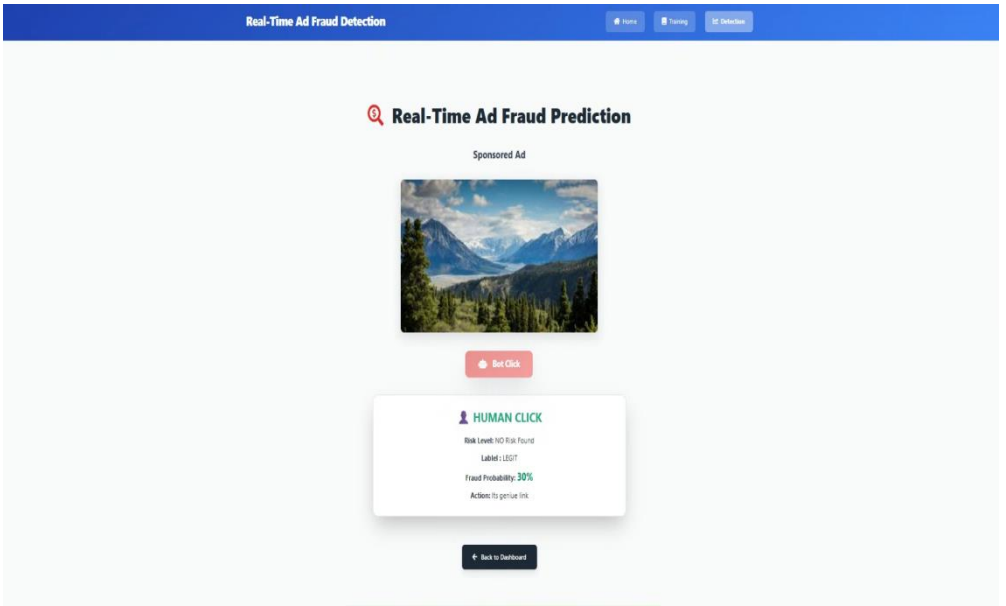

Figure 5.2:Human click detection image of Ad Click
FraudDetection Using Machine Learning And Deep Learning Algorithms

5.3 Bot Click Detection :

In the below screen, when the user clicks on the advertisement image, the system analyzes the click and predicts it as a Bot Click, indicating fraudulent or automated activity.

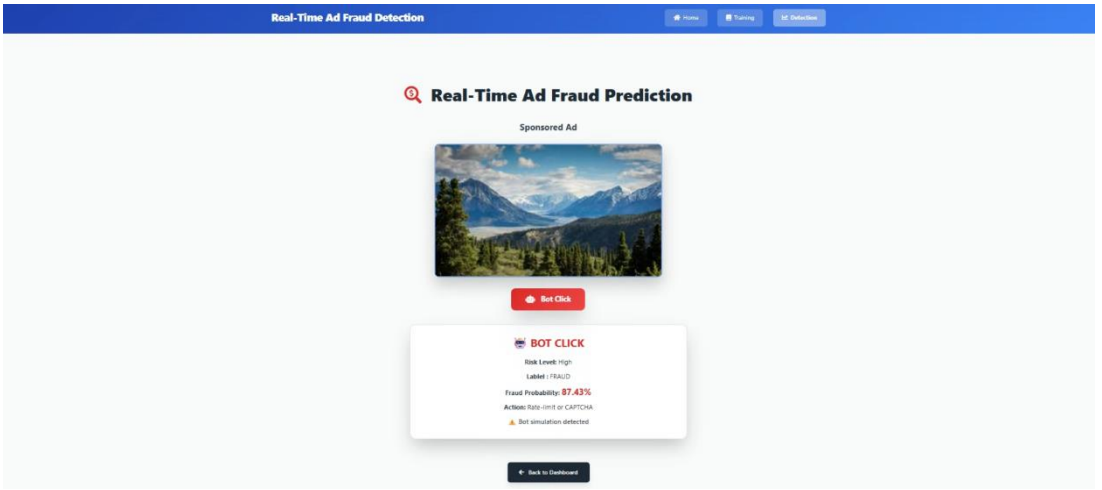

Figure 5.3: Bot click detection image of Ad Click
FraudDetection Using Machine Learning And Deep Learning Algorithms

5.4 Multiple Click Detection :

In the below screen, when a human user clicks the advertisement multiple times, the system analyzes the behavior and may classify it as fraudulent activity based on repeated click patterns.

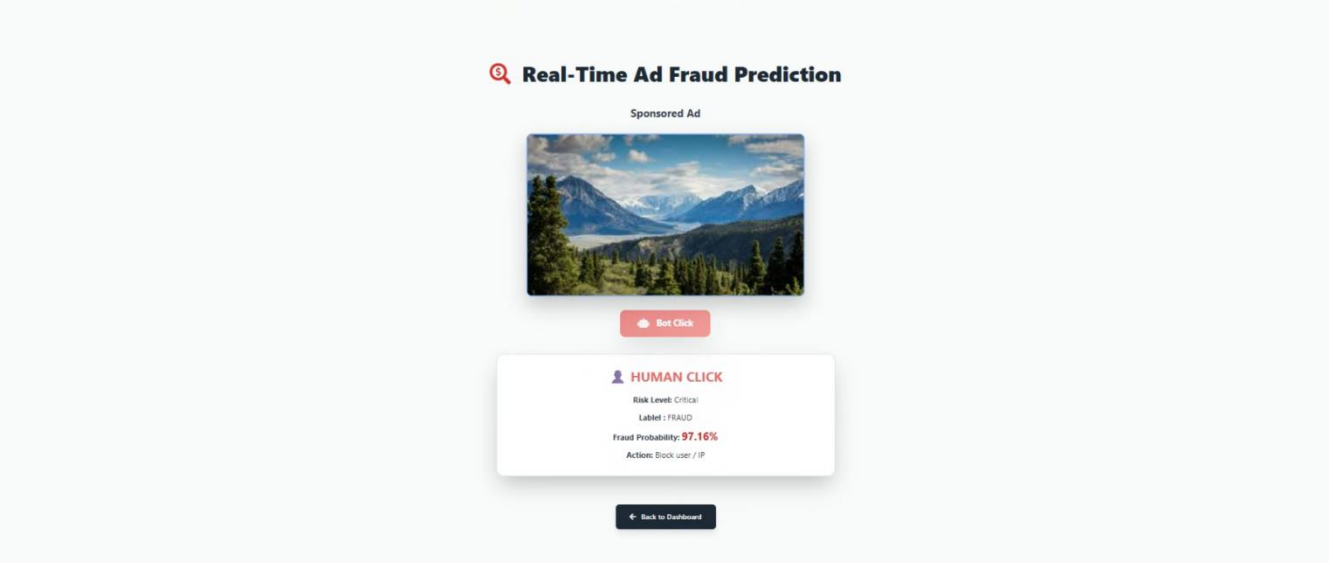

Figure 5.4: Multiple click detection image of Ad Click FraudDetection Using Machine Learning And Deep Learning Algorithms

5.5 Model Training :

In the below screen, when the user clicks on Start Training, the system trains the model using algorithms like Random Forest, LSTM, and Hybrid Model, and displays the training accuracy and performance results.

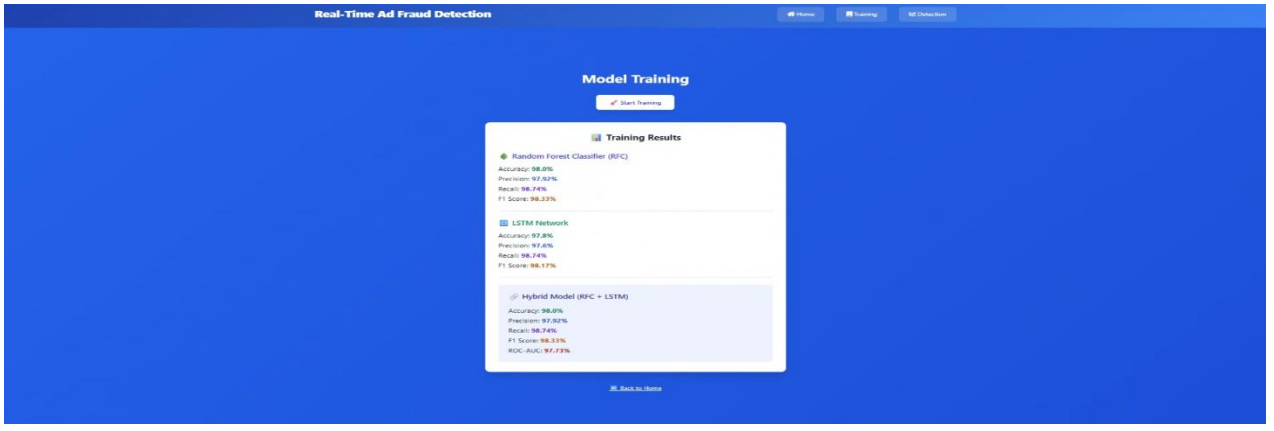

Figure 5.5.:Model Training image of Ad Click FraudDetection Using Machine Learning And Deep Learning Algorithms

6. VALIDATION

6. VALIDATION

6.1 INTRODUCTION

Validation of the proposed system was performed using the **test dataset**, which contains both legitimate and fraudulent click samples. After training the models using the training dataset, the system was evaluated on unseen data to measure its prediction capability.

Different evaluation metrics were used to validate the performance of the proposed fraud detection system:

1. Accuracy

Accuracy measures the percentage of correctly classified clicks (both fraudulent and legitimate) out of the total number of clicks.

2. Precision

Precision indicates how many clicks predicted as fraudulent were actually fraudulent. High precision means fewer false fraud detections.

3. Recall

Recall measures how many actual fraudulent clicks were successfully detected by the system.

4. F1 Score

F1 Score is the harmonic mean of precision and recall, providing a balanced evaluation of the model performance.

The Random Forest model provided strong performance for structured click data, while the LSTM model improved detection of sequential fraud patterns. Combining both approaches improves the overall effectiveness of the fraud detection system.

The validation results confirm that the proposed system can detect abnormal click behavior and classify fraudulent activities with high reliability. This system can be deployed in digital advertising platforms to monitor click activity in real time and reduce the impact of fraudulent clicks.

6.2 TEST CASES

TABLE 6.2.1 UPLOADING DATASET

Test case ID	Test case name	Purpose	Test Case	Output
1	User uploads Dataset.	Use dataset for training the model	The user uploads the dataset required for training the fraud detection model	Dataset successfully loaded
2	StartModel Train	Train the machine learning model	The user clicks on Start Training to train the model using dataset	Model training completed successfully
3	Detect Ad Click	Identify whether click is genuine or fraudulent	The user clicks on the advertisement image to start detection	System predicts Human Click or Bot Click
4	Human Click Detection	Identify legitimate click activity	The user clicks the ad image once	System displays Human Click – Legitimate activity
5	Bot Click Detection	Detect automated or fraudulent clicks	The system analyzes suspicious clicking pattern	System displays Bot Click – Fraud detected
6	Multiple Click Detection	Detect repeated clicking behavior	The user clicks the advertisement multiple times	System flags the activity as Fraudulent

7	View Training Results	Display model performance	After training, the system shows accuracy, precision, recall and F1 score	Training results displayed successfully

TABLE 6.2.2 CLASSIFICATION

Test case ID	Test case name	Purpose	Input	Output
1	Classification test 1	To check if the classifier correctly identifies legitimate clicks	Normal ad click by user	Human Click (Legitimate)..
2	Classification test 2	To check if the classifier detects fraudulent clicks	Automated or suspicious ad click	Bot Click (Fraud)
3	Classification test 3	To verify model prediction accuracy	Advertisement image is clicked once	Human Click detected
4	Classification test 4	To detect repeated clicking activity	User clicks the advertisement multiple times	Fraudulent activity detected
5	Classification test 5	To test bot simulation detection	System receives automated click pattern	Bot Click detected

7. CONCLUSION &FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

In conclusion, the project has successfully achieved its objectives, showcasing significant progress and outcomes. The implementation and execution phases were meticulously planned and executed, leading to substantial improvements and insights. Looking ahead, the future aspects of the project hold immense potential. Future developments will focus on expanding the scope, integrating new technologies, and enhancing sustainability. These advancements will not only strengthen the existing framework but also open new avenues for growth and innovation, ensuring the project remains relevant and impactful in the long term. This strategic approach will drive continuous improvement and success.

7.1 PROJECT CONCLUSION

The proposed Adaptive Hybrid Framework for Real-Time Ad Click Fraud Detection provides an efficient and reliable solution to identify fraudulent activities in online advertising. By combining machine learning models with LSTM-based deep learning, the system is able to analyze both behavioral patterns and temporal user interactions. This hybrid approach improves detection accuracy and reduces false positives and false negatives compared to traditional single-model methods.

The system supports real-time monitoring, making it suitable for large-scale advertising platforms where immediate fraud detection is essential. The integration of feature engineering, behavioral analysis, and sequential modeling enhances the system’s ability to detect complex and evolving fraud patterns. Overall, the proposed framework improves the security, transparency, and effectiveness of digital advertising systems.

7.2 FUTURE ASPECTS

Although the proposed system achieves improved performance, several enhancements can be considered in the future:

- Implementation of real-time streaming frameworks for large-scale deployment
- Integration of advanced deep learning models such as Transformer-based architectures
- Use of federated learning to improve privacy and distributed training
- Development of automated model update mechanisms for continuous learning
- Incorporation of additional behavioral and network-level features
- Deployment as a cloud-based service for scalable and flexible access
- Visualization dashboards for better monitoring and decision support

These improvements can further enhance the adaptability, scalability, and robustness of the click fraud detection system in real-world environments.

8. BIBLIOGRAPHY

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8.1 REFERENCES

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8.2 GITHUB LINK

https://github.com/patelnaveen7/AD-CLICK-FRAUD-DETECTION-USING-MACHINE-LEARNING-AND-DEEP-LEARNING-ALGORITHMS